GOVERNMENT POLYTECHNIC, KARAD

EVALUATION SHEET FOR MICRO PROJECT

Name of Programme: Electrical Engineering **Semester:** EE5I

Course Title :Maintenance of Electrical Equipment **Code:** 22625

Title of Micro-Project : To prepare a performance model of Single Phase Induction Motor.

Course Outcomes Achieved:

Major Learning Outcomes achieved by student while doing the Project:

a) Unit Outcomes in Cognitive Domain:

1. Explain the steps in preparing trouble shooting chart for the given type of rotating machine.
2. Suggest the tools for the given operation under maintenance of the rotating machine.

b) Practical Outcomes :

1. Perform the no load test on , measure winding resistance for given single phase induction motor and determine its performance.

c) Outcome in Affective Domain:

1. Follow ethical & safety practices.
2. Demonstrate working as a leader/ a team member.

Comments / Suggestions about team work / leadership / inter-personal communication (if any):

Roll Numbers	Student Name	Marks out of 6 For Performance in Group activity (D5 Col.8	Marks out of 4 for performance in oral / Presentation (D5 Col. 9)	Total Marks (10 Marks)
2301	Gokhale Pratikraj Vinayak			
2302	Lohar Anurag Santosh			
2303	Chavan Soham Sanjay			
2304	Kumbhar Tushar Gajendra			
2305	Jadhav Gouri Shahaji			
2306	Shinde Pranali Ramdas			
2307	Jadhav Surya Sudhir			
2308	Patil Sourabh Lakshaman			
2309	Deshmukh Bhagyashree Prakash			
2310	Sawant Swarali Sharad			
2311	Gorad Vishnu Prathmesh			
2312	Jadhav Anuj Pravin			

2313	Pawar Pranali Pradip			
2314	Thorat Srushti Deepak			
2315	Bhoye Dhiraj Vijay			
2316	Kambale Sangharsh Sameer			
2317	Kumbhar Swaranjali Vasant			
2318	Shinde Tanvi Mahesh			
2319	Shinde Sanjana Ganesh			
2320	Sutar Yash Prakash			
2321	Shinde Sakshi Sunil			
2322	Porlekar Devika Deepak			
2323	Sathe Roshani Jaykar			

Name & Designation of the Teacher: Prof.C.S.Bandgar

Dated Signature :

GOVERNMENT POLYTECHNIC, KARAD

Evaluation as per Rubrics for Assessment of Micro Project

Name of Programme: Electrical Engineering **Semester:** EE5I

Course Title : Maintenance of Electrical Equipment **Code:** 22625

Title of Micro-Project:. To prepare a performance model of Single Phase Induction Motor.

Sr. No	Roll Numbers :→	2301 to 2324			
(A) Process and Product Assessment (Convert total marks out of 6 Marks)					
1	Relevance to the course				
2	Literature Review / Information collection				
3	Completion of the target as per Project proposal				
4	Analysis of Data and representation				
5	Quality of Prototype / Mode				
6	Report Preparation				
Sub Total (Out of 60)					
Total Out of 6 Marks					
7	Presentation				
8	Viva				
Sub Total (Out of 20)					
Total Out of 4 Marks					
Total Marks (A+B)					

**Performance (Marks): Poor (1-3), Average (4-5), Good (6-8), Excellent (9-10).

Roll Numbers	Process & Product Assessment (6 Marks)	Individual Presentation / Viva (4 Marks)	Total Marks (10 Marks)
2301			
2302			
2303			
2304			
2305			
2306			
2307			
2308			

2309			
2310			
2311			
2312			
2313			
2314			
2315			
2316			
2317			
2318			
2319			
2320			
2321			
2322			
2323			

Name & Designation of the Teacher: Prof. C.S.Bandgar

Dated Signature :

A

Micro-Project Report

On

"To prepare a performance model of Single Phase Induction Motor."

Submitted By,

Roll Numbers	Student Name	Marks out of 6 For Performance in Group activity (D5 Col.8	Marks out of 4 for performance in oral / Presentation (D5 Col. 9)	Total Marks (10 Marks)
2301	Gokhale Pratikraj Vinayak			
2302	Lohar Anurag Santosh			
2303	Chavan Soham Sanjay			
2304	Kumbhar Tushar Gajendra			
2305	Jadhav Gouri Shahaji			
2306	Shinde Pranali Ramdas			
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2317	Kumbhar Swaranjali Vasant			
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2319	Shinde Sanjana Ganesh			
2320	Sutar Yash Prakash			
2321	Shinde Sakshi Sunil			
2322	Porlekar Devika Deepak			
2323	Sathe Roshani Jaykar			

Under the guidance of

Prof. C.S.Bandgar

Department of Electrical Engineering

Government Polytechnic, Karad

Even Semester 2024-25

Government Polytechnic, Karad

Department of Electrical Engineering

Certificate

This is to certify that,

Roll Numbers	Student Name	Marks out of 6 For Performance in Group activity (D5 Col.8	Marks out of 4 for performance in oral / Presentation (D5 Col. 9)	Total Marks (10 Marks)
2301	Gokhale Pratikraj Vinayak			
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2320	Sutar Yash Prakash			
2321	Shinde Sakshi Sunil			
2322	Porlekar Devika Deepak			
2323	Sathe Roshani Jaykar			

have successfully completed the micro project **on “To prepare a performance model of Single Phase Induction Motor.”** of the **Maintenance of Electrical Equipment (22625)** prescribed by Maharashtra State Board of Technical Education for Fifth Semester (Electrical) during the year 2024-25.

Prof. C.S.Bandgar

Project Guide

Department of Electrical Engineering

Government Polytechnic, Karad

Acknowledgement

We express our deep sense of gratitude towards Prof. for his consistent guidance and co-operation for the project. We like to thank him for his kind suggestions and support.

We are also thankful of Dr. S.A.Patil (Principal, Government Polytechnic, Karad) for his support.

We are grateful towards the supporting staff of Electrical Engineering Department, Government Polytechnic, Karad for providing us the necessary tools and facilities for our project work.

Finally, we are thankful to all those who have directly or indirectly helped us for this project. This project has given us great learning experience and sense of satisfaction.

Thanking You,

Gokhale Pratikraj Vinayak
Lohar Anurag Santosh
Chavan Soham Sanjay
Kumbhar Tushar Gajendra
Jadhav Gouri Shahaji
Shinde Pranali Ramdas
Jadhav Surya Sudhir
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Shinde Tanvi Mahesh
Shinde Sanjana Ganesh
Sutar Yash Prakash
Shinde Sakshi Sunil
Porlekar Devika Deepak
Sathe Roshani Jaykar

Micro-Project Proposal

"To prepare a performance model of Single Phase Induction Motor."

1.0 Aims / benefits of the Micro-Project:

1. To know the performance of Single Phase Induction Motor , by understanding its circuit parameters and behaviour in technical domain.

2. This model gives accessibility to measure all the required parameters to determine its efficiency , ratings and performance characteristics.

2.0 Course Outcomes Addressed:

1. Maintain rotating electrical machines .
2. Maintain insulation system of all electrical equipment**.**

3.0 Proposed methodology:

1. Assign the topic of Micro-Project.
2. Equal distribution of work among the all-group members.
3. Firstly, searched the material required for Single Phase Induction Motor performance model along with taking in account selection of proper rating of the component and economical view .
4. Study the given setup.
5. Made circuit connection as per requirement of parameters to be measured .
6. Verify the information from subject teacher.
7. Prepare a report on micro project subject.
8. At last, give report presentation to teacher and submit report.

4.0 Action Plan:

Sr. No	Details of activity	Planned Start Date	Planned Finish Date	Name of Team Members
1	Collection of different project topic	09/01/2025	16/01/2025	Shinde Pranali Ramdas Jadhav Gouri Shahaji
2	Selection of the project topic.	23/01/2025	29/01/2025	Gorad Prathmesh Vishnu Bhoye Dhiraj Vijay Kambale Sangharsh Jadhav Anuj Pravin
3	Collecting information related to given project./Actual Execution of project	07/01/2025	14/01/2025	Lohar Anurag Santosh Gokhale Pratikraj Kumbhar Tushar

4	Project proposal making	14/01/2025	21/01/2025	Porlekar Devika Deepak
5	Report Making	06/03/2025	13/03/2025	Pawar Pranali Pradip
6	Project report correction from teacher	03/04/2025	03/04/2025	All Team Members
7	Project submission			All Team Members

5.0 Resources Required:

Sr. No.	Name of Resources / material	Specification	Quantity	Remark
1	Capacitor Start Capacitor Run 1 ph Induction Motor	0.75kw/1 HP,1450 RPM,6.70 A ,Power factor :075, 100/120 µf capacitor	01	
2	Terminals (Red & Black)	Red= 3 Black= 5	08	
3	Wooden Board	-	01	
4	Tester/Screwdriver/Wire stripper	-	03	

Approved by –
Prof.Mrs C.S.Bandgar,
Faculty Guide,
Dept. of Electrical Engineering
G. P. Karad

A

Micro-Project Report

"To prepare a performance model of Single Phase Induction Motor."

1.0 Rationale :

This project offers an advantage of thoroughly knowing the performance of single phase capacitor start capacitor run induction motor . All the circuit parameters of the motor can be actually found and based on that it can be possible to determine motor losses, efficiency etc.

2.0 Aims / benefits of the Micro-Project:

1. To know the performance of Single Phase Induction Motor , by understanding its circuit parameters and behaviour in technical domain.

2. This model gives accessibility to measure all the required parameters to determine its efficiency , ratings and performance characteristics.

3.0 Course Outcomes Addressed:

1. Maintain rotating electrical machines .
2. Maintain insulation system of all electrical equipment**.**

4.0 Literature Review :

Introduction :

The capacitor start capacitor run motor, also known as the two-value capacitor motor, is a type of single-phase induction motor that uses two capacitors - a starting capacitor and a running capacitor. The starting capacitor provides an initial boost of energy needed for startup, while the running capacitor continues providing phase correction throughout the motor's operation. With its unique design incorporating two capacitors, this motor exhibits outstanding performance characteristics like high starting torque and improved efficiency, making it an ideal solution for applications requiring frequent starts or high starting loads.

Definition of a Capacitor Start Capacitor Run Motor

A capacitor start capacitor run motor is a type of single-phase induction motor that incorporates both a start capacitor and a run capacitor. These capacitors are used to create a phase shift in the motor's windings, improving its starting torque and running efficiency.

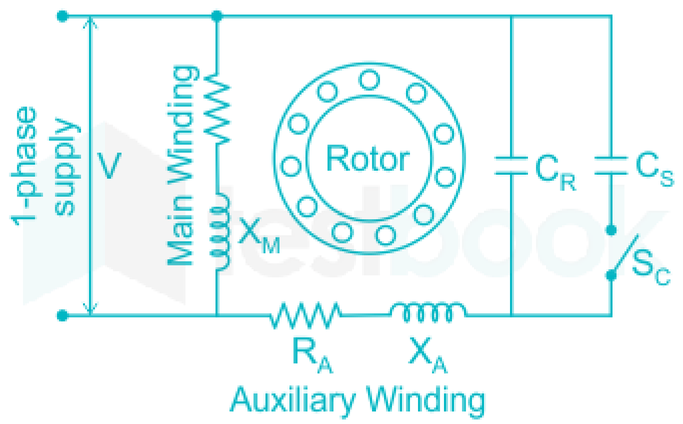
The capacitor start capacitor run motor features a cage rotor, with its stator comprising two windings: the Main Winding and the Auxiliary Winding. These windings are spatially displaced by 90 degrees. This motor employs two capacitors: the starting capacitor, which is utilized during startup to provide high initial torque, and the run capacitor, which is engaged for continuous operation, ensuring efficient and stable performance.
Key Components:

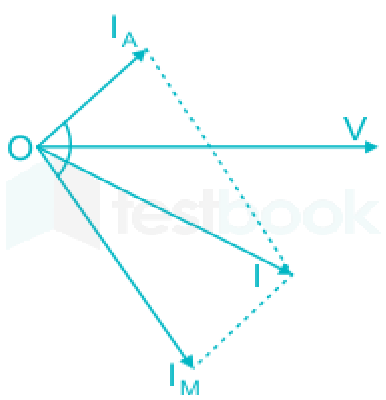
- **Start Capacitor:** Provides a high starting torque by creating a large phase shift during the start-up.
- **Run Capacitor:** Maintains the phase shift during normal operation, ensuring smooth and efficient running.

Working of a Capacitor Start Capacitor Run Motor:

The working principle of the capacitor start capacitor run motor relies on creating a rotating magnetic field using phase correction provided by the capacitors.
At startup, the starting capacitor (Cs) connected in series with the auxiliary winding generates a leading current which is 90° ahead of the main winding current. This phase difference is essential for producing the rotating magnetic field needed to develop torque and get the rotor rotating.
As the motor reaches nearly synchronous speed, the starting capacitor is cut off automatically through a centrifugal switch. Now only the running capacitor (Cr) connected in parallel to the auxiliary winding provides the necessary 90° lagging current for continued rotation.
The running capacitor being permanently connected improves the motor's power factor during steady-state operation. The continuous phase correction supplied by it allows smooth, vibration-free running of the motor.

This unique two-capacitor design enables the motor to produce high starting torque for quick acceleration and also high operating efficiency through a good power factor - making it suitable for heavy-duty applications requiring frequent starts.



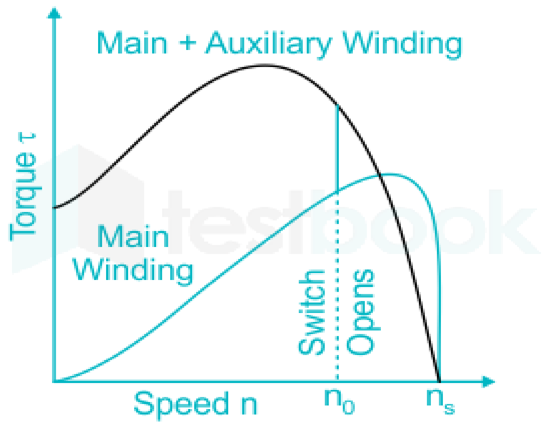


The torque-speed characteristic of a Capacitor Start Capacitor Run Motor

The torque-speed characteristic is an important parameter that determines a motor's performance for various applications. The torque-speed curve of a capacitor start capacitor run motor is as follows:

- At startup when both capacitors are engaged, the motor develops very high starting or breakaway torque sufficient for difficult loads.
- As the speed increases progressively, the torque decreases steadily along the curved line.
- Near synchronous speed, the starting capacitor drops out automatically but the torque is maintained at a constant level due to the running capacitor.
- In the constant torque region, the motor continues to deliver uniform torque output without any torque pulsations even at higher speeds.
- In the constant power region above the base speed, the torque falls gradually while the power output is maintained constant up to the maximum or rated speed.

This torque behavior ensures smooth, jerk-free acceleration and operation of loads driven by the capacitor start capacitor run motor.



Advantages of Capacitor Start Capacitor Run Motor

Some of the key advantages of this special electric motor include:

- High starting/breakaway torque for easy starts under full load conditions
- Improved power factor and higher efficiency due to permanent phase correction from the running capacitor
- Constant torque characteristic leads to smooth, pulsation-free operation
- Compact and lightweight design suitable for integration with different machine tools.
- Quiet running performance is useful in settings requiring low noise levels.
- Long mechanical life due to negligible starting and stopping stresses on components.
- Operates at higher efficiencies compared to conventional induction motors.
- Maintains uniform speed under varying loads.
- Self-starting without the need for external starters.
- Low cost of installation and maintenance.

Disadvantages of Capacitor Start Capacitor Run Motor

Some potential downsides of this motor type are:

- Higher initial cost compared to shaded-pole or split-phase induction motors
- Prone to failure of capacitors requiring periodic replacement
- Complex internal construction with additional switching components
- Difficulty in reversing the direction of rotation without rewiring
- Lower accelerating torque than a split-phase motor of same rating
- Requires careful sizing of capacitors for optimal performance
- Higher losses due to frictional dissipation in capacitor switch

Thus while offering distinct advantages, the capacitor start capacitor run motor demands more careful sizing, installation practices, and regular maintenance for reliable long-term operation.

5.0 Actual Methodology Followed :

1. Assign the topic of Micro-Project.
2. Equal distribution of work among the all-group members.
3. Firstly, searched the material required for Single Phase Induction Motor performance model along with taking in account selection of proper rating of the component and economical view .
4. Study the given setup.
5. Made circuit connection as per requirement of parameters to be measured .
6. Verify the information from subject teacher.
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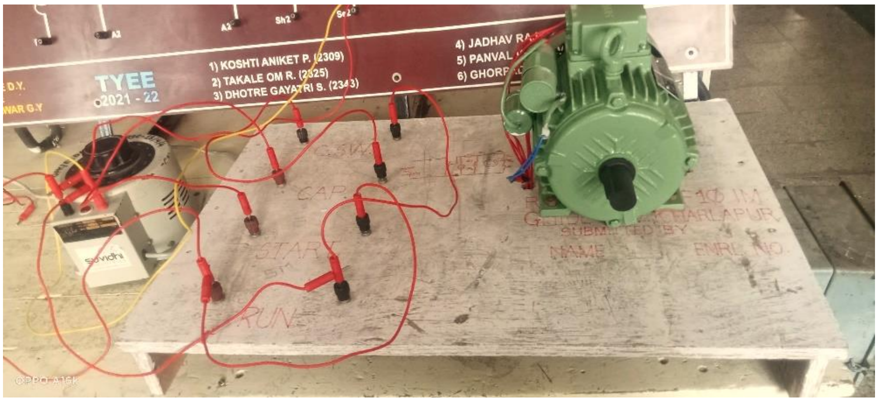
6.0 Resources Required:

Sr. No.	Name of Resources / material	Specification	Quantity	Remark
1	Capacitor Start Capacitor Run 1 ph Induction Motor	0.75kw/1 HP,1450 RPM, 6.70 A,Power factor :075, 100/120 µf capacitor	01	
2	Terminals (Red & Black)	Red= 3 Black= 5	08	
3	Wooden Board	-	01	
4	Tester/Screwdriver/Wire stripper	-	03	
5				

7.0 Output of the Micro-Project:



Performance model for Capacitor Start Capacitor Run Induction Motor

Procedure: -

1.Connect the circuit as per the circuit diagram.

2. To conduct no load test apply rated voltage to the motor using 1-Φvariac.

3.Note the readings of Voltmeter, Ammeter & Wattmeter.

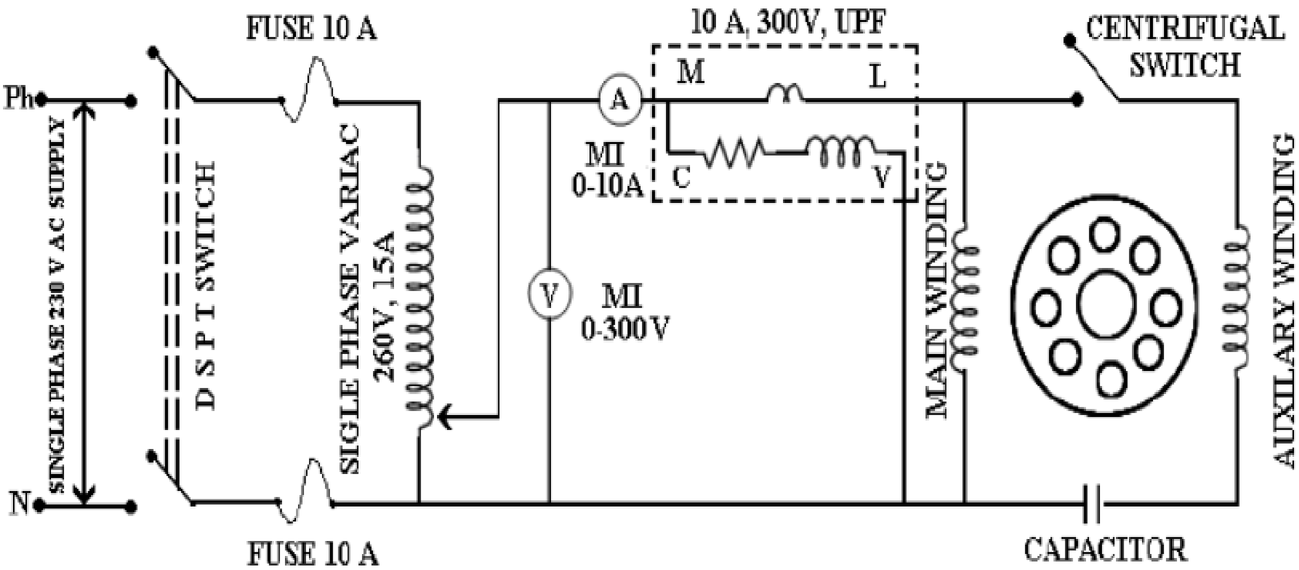
4.Now block the rotor to conduct blocked rotor test.

5.Apply voltage to the motor slowly using variac such that ratedcurrent flows in the stator.

6.Note down the readings of Voltmeter, Ammeter & Wattmeter.

7.Calculate the motor parameters and draw the equivalent circuit.

Circuit Diagram: -



NO-LOAD & BLOCKED ROTOR TEST ON 1-Φ INDUCTION MOTOR CIRCUIT DIAGRAM

Observation & Calculations:

Sr.No.	Winding	Resistance (in ohm)	Identification
1	Primary Winding	11 ohm	Starting Winding
2	Secondary Winding	4 ohm	Main Winding

No Load Test :

Sr.No.	V_0	I_0	W_0	Speed (rpm)
1	200 V	2 A	40 W	1450 rpm

8.0 Skill Developed / Learning Outcomes of this Micro-project:

1. Group Working Skills
2. Leadership Skills
3. Ethics

9.0 Applications of this Micro-Project:

Given its beneficial performance traits, the capacitor start capacitor run induction motor finds usage across many industries for driving applications involving frequent starts, high starting loads, or the need for smooth, constant speed operation. Some common applications include:

1. Pumps (vane, centrifugal, gear) used in irrigation, water supply, sewage, etc.
2. Fans and blowers in ventilation, and air conditioning plants
3. Screw compressors in refrigeration, air conditioning units
4. Hoists, cranes, and elevators for frequent load handling
5. Conveyors and material handling systems in process plants
6. Machine tools like lathes, mills, and drilling machines.